

Calculus Skeleton Notes

Contents

September 9: Average Rates of Change	1
September 11: 1.1 - Limits	3
September 15: 1.3 - Finding Limits Analytically	7
September 16: 1.5 - Continuity	11
September 18: 1.6 - Limits Involving Infinity	15
September 22: 2.1 - Derivatives	18
September 23: 2.2 - Derivatives	21
September 25: 2.3 - Differentiation Rules	23
September 29: 2.4 - Product and Quotient Rule	27
October 2: 2.5 - Chain Rule	31
October 6: 2.6 - Implicit Differentiation	34
October 9: 2.7 - Inverses and Inverse Trig	37

Average Rates of Change

Goals: Compute average rates of change, slopes of secant lines, and motivate the idea of limits

Before Class Videos: None today, starting next class

Example 1: I left my house at time $t = 0$. 30 minutes later, I was at UVic which is 12 km away.

a) What was my average speed?

b) Sketch a possible graph

c) How could I approximate my speed at $t = 10$ minutes?

Example 2: What is average rate of change of $f(x) = x^2$ on $[2, 4]$?

Example 3: Find the equation of the line through (x_1, y_1) and (x_2, y_2) .

Example 4: Consider the graph of a function $f(x)$. How do I write an equation for the secant line through $(x_1, f(x_1))$ and $(x_2, f(x_2))$?

Big Idea: As x_2 gets closer to x_1 the slope of the secant line approaches the slope of the tangent line at x_1 . We need to investigate and define this limiting process more precisely. This DESMOS visualizes the idea: <https://www.desmos.com/calculator/oi7poyhodr>

Suggested after class problems from APEX: None today, but review the Course Outline on Brightspace

1.1 - Limits

Goals: Describe limits graphically, numerically, and algebraically.

Before Class Videos: 1.1.1 and 1.1.2

Example 1: Consider these three functions:

$$f(x) = x + 1, \quad g(x) = \frac{x^2 - 1}{x - 1} = \frac{(x + 1)(x - 1)}{x - 1}, \quad h(x) = \begin{cases} x + 1, & x \neq 1, \\ 3, & x = 1. \end{cases}$$

How are these similar or different?

Definition:

$\lim_{x \rightarrow a} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently close to a .

Example 2: Compute the limit as $x \rightarrow 1$ of $f(x)$, $g(x)$, and $h(x)$.

Example 3: Use a table of values to estimate $\lim_{x \rightarrow 1} (x + 1)$.

Example 4: Use a table of values to estimate $\lim_{x \rightarrow 0} \frac{1}{x}$.

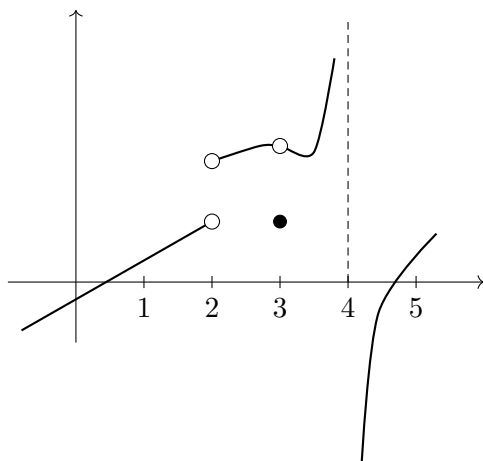
Example 5: Investigate

$$f(x) = \begin{cases} x, & x < 1, \\ x + 1, & x \geq 1. \end{cases}$$

Definition:

$\lim_{x \rightarrow a^+} f(x)$ means $f(x)$ gets arbitrarily close to L for all x sufficiently close to a from the right,
i.e. $x > a$.

Example 6: Use the graph to evaluate the indicated limits.



$$\lim_{x \rightarrow 2^-} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 2^+} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 2} f(x) = \underline{\hspace{2cm}},$$

$$\lim_{x \rightarrow 3^-} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 3^+} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 3} f(x) = \underline{\hspace{2cm}},$$

$$\lim_{x \rightarrow 4^-} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 4^+} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 4} f(x) = \underline{\hspace{2cm}}.$$

Example 7: Investigate limits of

$$f(x) = \sin\left(\frac{1}{x}\right)$$

Example 8: Graphically investigate limits of $\frac{x^2-1}{x^2-x-2}$

Suggested after class problems from APEX: Section 1.1 problems 9,11,15

1.3 - Finding Limits Analytically

Goals: Evaluate limits using limit laws, algebra, and the Squeeze Theorem.

Before Class Videos: 1.3.1–1.3.3

Definition:

If $f(x)$ gets arbitrarily close to L for all x sufficiently close to a , then

$$\lim_{x \rightarrow a} f(x) = L.$$

For many “nice” functions, evaluating a limit just means plugging in the number. We’ll investigate what functions are “nice” in section 1.5 on Continuity. For example:

Example 1:

- $\lim_{x \rightarrow 2} x^3 =$
- $\lim_{x \rightarrow \pi} \sin(x) =$
- $\lim_{x \rightarrow e} \ln(x) =$
- $\lim_{x \rightarrow 4} \sqrt{x} =$

Theorem:

If L , M , c , and k are real numbers and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M,$$

then

- | | |
|----------------------------|---|
| 1. Sum Rule: | $\lim_{x \rightarrow c} [f(x) + g(x)] = L + M$ |
| 2. Difference Rule: | $\lim_{x \rightarrow c} [f(x) - g(x)] = L - M$ |
| 3. Constant Multiple Rule: | $\lim_{x \rightarrow c} [kf(x)] = kL$ |
| 4. Product Rule: | $\lim_{x \rightarrow c} [f(x)g(x)] = LM$ |
| 5. Quotient Rule: | $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$ |
| 6. Power Rule: | $\lim_{x \rightarrow c} [f(x)]^n = L^n, \quad n \text{ a positive integer}$ |
| 7. Root Rule: | $\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, \quad n \text{ a positive integer}$ |

If n is even in the Root Rule, assume that $f(x) \geq 0$ on an interval containing c .

Example 2: Evaluate

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x + 2}.$$

Example 3: Come up with functions f and g such that

$$\lim_{x \rightarrow 3} f(x)g(x) \neq \left[\lim_{x \rightarrow 3} f(x) \right] \left[\lim_{x \rightarrow 3} g(x) \right].$$

Four Algebraic Tricks

Example 4:

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x - 2}.$$

Example 5:

$$\lim_{x \rightarrow 0} \frac{(x - 2)^2 - 4}{x}.$$

Example 6:

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}.$$

Example 7:

$$\lim_{x \rightarrow 2} \frac{\frac{1}{2} - \frac{1}{x}}{x - 2}.$$

Example 8: Evaluate

$$\lim_{x \rightarrow 4} \frac{4 - x}{5 - \sqrt{x^2 + 9}}.$$

Theorem:

Squeeze Theorem. If $f(x) \leq g(x) \leq h(x)$ for all x in an interval containing a (except possibly at a), and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L,$$

then

$$\lim_{x \rightarrow a} g(x) = L.$$

Example 9: Evaluate

$$\lim_{x \rightarrow 0} x^2 \sin(1/x^2).$$

Example 10: Suppose

$$-x^2 + 4x + 1 \leq g(x) \leq -2x + 10$$

for all x . For what value of a can you determine $\lim_{x \rightarrow a} g(x)$? What is the limit?

1.5 - Continuity

Goals: Determine where a function is continuous, build continuous piecewise functions, and apply the Intermediate Value Theorem.

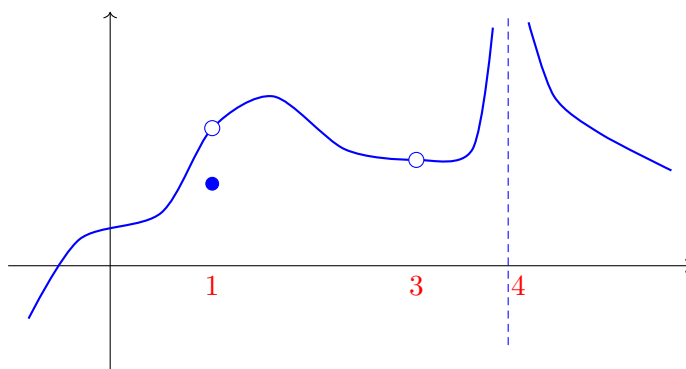
Before Class Videos: Section 1.5 videos 1 and 3

Definition:

A function f is **continuous at** c if all three of the following hold:

1. $f(c)$ exists,
2. $\lim_{x \rightarrow c} f(x)$ exists, and
3. $\lim_{x \rightarrow c} f(x) = f(c)$.

Example 1: Identify the type of discontinuity at each marked x -value.



Example 2: For what values of a is

$$f(x) = \begin{cases} x + 2, & x \leq 1, \\ ax^2, & x > 1 \end{cases}$$

continuous everywhere?

Example 3: For what values of a and b is

$$f(x) = \begin{cases} -2, & x \leq -1, \\ ax - b, & -1 < x < 1, \\ 3, & x \geq 1 \end{cases}$$

continuous everywhere?

Combining Continuous Functions

Theorem:

If f and g are continuous at c , then $f + g$, $f - g$, constant multiples of f , and fg are continuous at c . The quotient f/g is also continuous at c provided $g(c) \neq 0$.

Let's justify the sum rule from the limit laws:

Composition of Functions

Example 4: Compute the domain and range of $\sin \sqrt{x}$

Theorem:

Suppose f is continuous at a and g is continuous at $f(a)$. Then the composition $g \circ f$ is continuous at a , and

$$\lim_{x \rightarrow a} g(f(x)) = g(f(a)).$$

Example 5: On what intervals is $\sin\left(\frac{x-2}{x^2+5x+7}\right)$ continuous?

Example 6: Evaluate

$$\lim_{x \rightarrow 1} \frac{\sin(\pi x/6)e^{x^3} + x}{x^2 + 2}.$$

Theorem:

Intermediate Value Theorem. If f is continuous on $[a, b]$ and y_0 is between $f(a)$ and $f(b)$, then there is some $c \in [a, b]$ such that

$$f(c) = y_0.$$

Sketch:

Example 7: Sketch examples where the assumptions of the Intermediate Value Theorem are not satisfied.

Example 8: Show that

$$x^3 - 3x - 1 = 0$$

has a solution in $[0, 2]$.

Example 9: Use the intermediate value theorem to show that at some time a baby's height equalled their child's height.

Suggested after class problems from APEX: Section 1.5 problems 21,24,35

1.6 - Limits Involving Infinity

Goals: Investigate the special $\sin(x)/x$ limit, compute limits whose values are infinity, as well as limits at infinity

Before Class Videos: 1.6.1-1.6.3

Example 1: Explore $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

Example 2: Compute $\lim_{x \rightarrow 0} \frac{\sin(5x)}{\sin(3x)}$.

Example 3: Explore

$$f(x) = \frac{x}{x-2}$$

and find all asymptotes.

Definition:

$\lim_{x \rightarrow a} f(x) = \infty$ if $f(x)$ can be made arbitrarily large for all values of x sufficiently close to a .

Note: A limit being infinite is a particular example of a limit not existing.

Definition:

$\lim_{x \rightarrow \infty} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently large.

Example 4:

$$\lim_{x \rightarrow \infty} \sin(1/x)$$

Example 5:

$$\lim_{x \rightarrow \infty} x \sin(1/x)$$

Example 6: Find all Horizontal Asymptotes for

$$\frac{2x^3 + 3x}{x^4 - x^2}.$$

Example 7: Find all Horizontal Asymptotes for

$$\frac{\sqrt{x^6 + 2x}}{x^3 + x^{1/3}}.$$

Example 8: Find VA's; HA for $e^{1/x}$; graph it!

Example 9: Give example of a function with two HAs, $y = 0$ and $y = 2$, continuous everywhere.
Hint: $\arctan(x)$.

Suggested after class problems from APEX: Section 1.4 #40 and Section 1.6 #10, 19, 27

2.1 - Derivatives

Goals: Define the Derivative and apply it to initial examples

Before Class Videos: 2.1.1, 2.1.2

Example 1: Recall the velocity problem from the first day of class. If Trefor has some distance function $f(t)$, how fast is he going at 10 minutes? Let's explore graphically and numerically:

Definition:

The derivative of $f(x)$ at c is given by

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

provided this limit exists. If it does, we say f is differentiable at c .

Example 2: Compute the derivative of $f(x) = x^2$ at $x = 2$.

Definition:

The derivative of $f(x)$ on an open interval I is the function $f'(x)$ given by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists for all x in I . If it does we say $f(x)$ is differentiable on I .

Example 3: $f(x) = 1/x^2$.

Example 4: $f(x) = \sqrt{x}$.

Example 5: Let's explore if the following functions are differentiable at $x = 0$:

a)

$$f(x) = \begin{cases} 2, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

b) $f(x) = |x|$

c) $f(x) = x^{2/3}$

d) $f(x) = x^{1/3}$

2.2 - Derivatives

Goals: A shorter than normal worksheet with a few extras on derivatives

Before Class Videos: Optional tricky example from last class's content: 2.1.4

Definition:

The derivative of $f(x)$ at c is given by

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

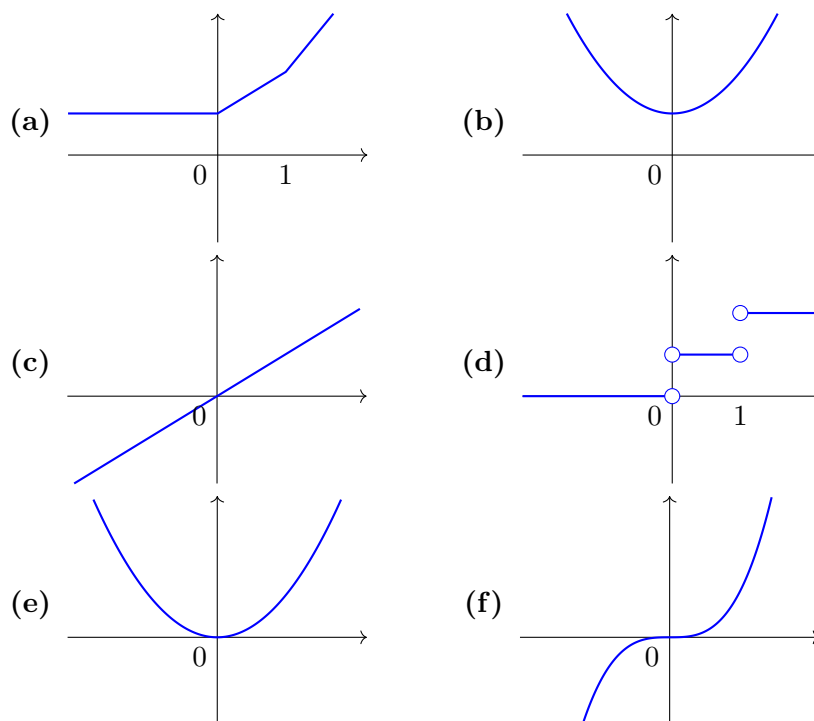
provided this limit exists. If it does, we say f is differentiable at c .

Definition:

An alternate formula for the derivative of $f(x)$ at c is

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}.$$

Example 1: The following six graphs consists of three pairs of functions and their derivative. Match them up.



Theorem:

Differentiable $\implies$ Continuous.

Note: This worksheet is deliberately short. I anticipate this class being a bit behind and finishing up prior material. If I am wrong I will edit it and add to it. The next worksheet is a more normal length.

Suggested after class problems from APEX: Section 2.2 15,17

2.3 - Differentiation Rules

Goals: State and apply Derivative Rules

Before Class Videos: 2.3.1

Theorem (Power Rule):

- $\frac{d}{dx}(c) = 0$ where c is a constant.
- $\frac{d}{dx}(x^n) = nx^{n-1}$ where $n \neq 0$

Theorem:

If $f(x)$ and $g(x)$ are differentiable at $x = a$, then

1. Sum/Difference Rule:

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$$

2. Constant Multiple Rule:

$$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$$

Example 1: Compute the derivative of $f(x) = 3x^4 - 5\sqrt{x} + 6x^{3/2} - \frac{7}{x^2} + 1$.

Proof of Theorem:

Definition:

The Point-Slope formula for the equation of the straight line through (x_0, y_0) with slope m is

$$y = y_0 + m(x - x_0)$$

Example 2: Find the equation of tangent line to a differentiable function $f(x)$ at the point x_0 .

Example 3: Find the equation of the tangent line to the function $f(x) = 2x^2 + 1$ at the point $(1, 3)$.

Common Derivatives:

Example 4: Explore the graphs of the following functions to discover their derivatives.

- $\frac{d}{dx}(e^x) =$

- $\frac{d}{dx} \ln(x) =$ for $x > 0$

- $\frac{d}{dx} \sin(x) =$

- $\frac{d}{dx} \cos(x) =$

Higher Derivatives: The derivative $f'(x)$ is itself a function we can take the derivative of. We call this the second derivative and might denote it by

$$f''(x) \quad \text{or} \quad \frac{d^2}{dx^2} f(x).$$

Example 5: Compute $f''(x)$ when $f(x) = e^x + \ln(x)$.

Example 6: If $f(x) = \sin(x)$ compute

- $f'(x) =$

- $f''(x) =$

- $f'''(x) =$

- $f^{(4)}(x) =$

- $f^{(2026)}(x) =$

Example 7: If $d(t)$ represents the position of a particle as a function of time, interpret $d'(t)$ and $d''(t)$.

Example 8: If $v(t)$ represents Trefor's total YouTube views after t days, interpret $v'(t)$ and $v''(t)$.

Suggested after class problems from APEX: Section 2.3 16, 21, 34 (tangent only)

2.4 - Product and Quotient Rule

Goals: Compute derivatives using the product and quotient Rules, including of trigonometric functions

Before Class Videos: 2.4.1 and 2.4.2

Theorem (Derivative Rules):

- $\frac{d}{dx}(c) = 0$ where c is a constant.
- $\frac{d}{dx}(x^n) = nx^{n-1}$ where $n > 0$ is an integer.

If $f(x)$ and $g(x)$ are differentiable at $x = a$, then

- $\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$
- $\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$

Let's add two new rules to the list

Theorem (Product and Quotient Rules):

$$\begin{aligned}\frac{d}{dx}[fg] &= f'g + fg' \\ \frac{d}{dx}\left[\frac{f}{g}\right] &= \frac{f'g - fg'}{g^2}\end{aligned}$$

Example 1: Compute the following derivatives

a)

$$\sqrt{x}e^x$$

b)

$$\frac{\sin(x)}{x^3}$$

c)

$$\frac{xe^x}{x^2 + 1}$$

Example 2: Let's explore the intuition behind why the product rule is NOT just the product of the derivatives.

Example 3: Let's finally figure out what exactly e^x is all about. We'll use the definition of the derivative to compute, for $a > 0$,

$$\frac{d}{dx}a^x$$

Let's explore

$$\lim_{h \rightarrow 0} \frac{a^h - 1}{h}$$

Definition:

The number e is defined to be the value of a such that $\lim_{h \rightarrow 0} \frac{a^h - 1}{h}$

Right-Triangle Definitions

$$\begin{array}{lll} \sin x = \frac{\text{opposite}}{\text{hypotenuse}} & \tan x = \frac{\text{opposite}}{\text{adjacent}} & \sec x = \frac{\text{hypotenuse}}{\text{adjacent}} \\ \cos x = \frac{\text{adjacent}}{\text{hypotenuse}} & \cot x = \frac{\text{adjacent}}{\text{opposite}} & \csc x = \frac{\text{hypotenuse}}{\text{opposite}} \end{array}$$

Rewriting with Sine and Cosine

$$\begin{array}{lll} \sin x = \sin x & \tan x = \frac{\sin x}{\cos x} & \sec x = \frac{1}{\cos x} \\ \cos x = \cos x & \cot x = \frac{\cos x}{\sin x} & \csc x = \frac{1}{\sin x} \end{array}$$

Example 4: Let's explore the graphs of these functions.

Example 5: Compute $\frac{d}{dx} \tan(x)$

Example 6: Compute $\frac{d}{dx} \sec(x)$

Trigonometric Derivatives

$$\begin{array}{lll} \frac{d}{dx} \sin x = \cos x & \frac{d}{dx} \tan x = \sec^2 x & \frac{d}{dx} \sec x = \sec x \tan x \\ \frac{d}{dx} \cos x = -\sin x & \frac{d}{dx} \cot x = -\csc^2 x & \frac{d}{dx} \csc x = -\csc x \cot x \end{array}$$

Suggested after class problems from APEX: Section 2.4 7, 18, 26

2.5 - Chain Rule

Goals: Compute the derivative of compositions of functions

Before Class Videos: 2.5.1, 2.5.2

Example 1: Consider $\sin(x^2)$. This is an example of a composition $f(g(x))$ or alternatively $f \circ g$.

Example 2: $f(x) = x \sin x$, $g(x) = \cos x + 1$. Write out:

- $f(g(x)) =$

- $g(f(x)) =$

- $f(f(x)) =$

- $g(g(x)) =$

Theorem (Chain Rule):

If g differentiable at x and f differentiable at $g(x)$ then

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x).$$

Example 3: $\frac{d}{dx} \sin(x^2 + 1)$

Example 4: $\frac{d}{dx} \sin^{30}(x)$

Example 5: $\frac{d}{dx} \cos(3e^x)$

Example 6: $\frac{d}{dx} \frac{1}{2x+1}$

Example 7: $\frac{d}{dx} \cos^2(x^3)$

Example 8: $\frac{d}{dx} \tan(x^2 e^x).$

Leibniz Notation: let $u = g(x)$,

$$\frac{d}{dx}[f(g(x))] = \frac{df}{du} \frac{du}{dx}.$$

Example 9: Car uses $6L/100km$. Cost is $1.5\$/L$. What is the rate of change in cost per $100km$?

Example 10: Gravel is dumped from a conveyor to form pile with

$$V(h) = \frac{\pi}{12}h^3.$$

At $5m$ tall, the pile is growing by 0.1 m/s. How fast is volume growing?

Example 11: Contrast graphically the slope of $f(x)$ and $f(2x)$ for an arbitrary function $f(x)$.

Example 12: Derive the Quotient Rule.

Suggested after class problems from APEX: Section 2.5: 7,17,24

2.6 - Implicit Differentiation

Goals: Compute derivatives of functions define implicitly

Before Class Videos: 2.6.1

Example 1: Explore $x^2 + y^2 = 1$. Where does it have vertical and horizontal tangents?

Big Idea: Treat $y = y(x)$ when x is independent and y is dependent.

Example 2: Find $\frac{dy}{dx}$ implicitly for $\sin(xy) + x = y^2 \cos x$.

Example 3: Compute $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = 4$.

Example 4: Now explore $x^3 + y^3 - 9xy = 0$. Find all vertical and horizontal tangents.

Logarithmic Differentiation: Recall we previously learned that $\frac{d}{dx} \ln(x) = \frac{1}{x}$.

Example 5: $y = x^x$.

Example 6:

$$y = \frac{(x+2)^3(x-1)^{1/3}}{(x+7)^2}.$$

Example 7:

$$y = x^{\cos x} \cdot (\cos x)^x.$$

2.7 - Inverses and Inverse Trig

Goals: Compute the Derivatives of Inverse functions and Inverse Trig functions.

Before Class Videos: 2.7.1

Note: This worksheet combined some content from inverses in general and trigonometric inverses. The former was covered on Oct 7th by Josh Manzer using his own worksheet and will thus not be covered extensively by Trefor, who will focus on the inverse trig for Friday's Class.

Definition:

For a function $f(x)$, let's define $f^{-1}(x)$ so

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x.$$

Example 1: $y = 2x + 1$

Example 2: $y = x^2$

Example 3: $\frac{d}{dx} \ln x$

Example 4: $\frac{d}{dx}a^x$

Example 5: Let's derive a general formula for the inverse of a function.

Example 6: $\tan^{-1}(\tan x) = x = \tan(\tan^{-1} x)$. Try evaluating these at $x = \pi$.

Example 7: $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Example 8: $\sin^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

Example 9: Plot the graph of $\arcsin(x)$ by comparing to the graph of $\sin(x)$.

Example 10: Compute $\frac{d}{dx} \arcsin(x)$.

Inverse Trigonometric Functions			
Function	Derivative	Domain	Range
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$\frac{1}{1+x^2}$	$\mathbb{R}$	$(-\pi/2, \pi/2)$
$\operatorname{arccot} x$	$-\frac{1}{1+x^2}$	$\mathbb{R}$	$(0, \pi)$
$\operatorname{arcsec} x$	$\frac{1}{ x \sqrt{x^2-1}}$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi/2) \cup (\pi/2, \pi]$
$\operatorname{arccsc} x$	$-\frac{1}{ x \sqrt{x^2-1}}$	$(-\infty, -1] \cup [1, \infty)$	$[-\pi/2, 0) \cup (0, \pi/2]$

The derivatives of $\arcsin x$ and $\arccos x$ hold for $|x| < 1$; those of $\operatorname{arcsec} x$ and $\operatorname{arccsc} x$ hold for $|x| > 1$.

Example 11: $\frac{d}{dx} \sin^{-1}(3x^2 + 1)$

Example 12: Let's do one more derivativion, this time of $y = \sec^{-1}(x)$.

Example 13: $\frac{d}{dx}[\ln(x^2 + 1) + x \tan^{-1}(2x)]$

Example 14: For $\tan^{-1} x + \sin^{-1} y = \frac{\pi}{4}$ find $\left. \frac{dy}{dx} \right|_{(0, \frac{1}{\sqrt{2}})}$